Supervisor Meeting Minutes
Meeting 4

Present: Matthew Chalmers, Jonas Leipner

Date: 20 October 2026

1 Achieved in Previous Week

Previous week: 13–20 October 2026

- Defined the game parameters that could be modified dynamically and separated them from functionality that should remain native and static.
- Considered how the dynamic execution model could be relevant beyond the game itself.

2 Issues Discussed & Solutions Agreed

- **Dynamic parameters:** Candidate values include tower damage, range, cooldown, targeting, enemy health, armour and behaviour, wave composition, gold rewards, incubator rules, and temporary balancing events.
- **Native parameters:** Flame rendering, lifecycle, input, collision detection, storage, permissions, and the validation bridge should remain native.
- **Execution flow:** The Flame loop passes combat state to a `CombatState` bridge, which sends data to `dart_eval`. The sandbox runs `calculateDamage()`, returns the result, and Flame validates and applies the damage.
- **Industry benchmarks:** Relevant examples include the dynamic compiler used by the PS2 version of Crash Bandicoot, DLL hot reloading, binary patching, and scripting VMs in games such as Skyrim and Roblox.
- **Broader relevance:** The project can be framed around dynamic loading in backend systems, plugin ecosystems, browser extensions, and recommender systems, rather than games alone.
- **Recommendation concept:** Telemetry recommendations could dynamically load UI widgets or features that help players discover effective configurations.

3 Objectives for This Week

Upcoming week: 20–27 October 2026

- Conduct a background survey of dynamic loading use cases.
- Design the initial game structure and telemetry recommendation framework.

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